EmbeddingGemma 2 on African Languages: Corrected Text Benchmarks and Speech Retrieval with a 5M-Parameter Adapter

Ugochukwu Onyebuchi
Carnegie Mellon University / NaijaML
pyvinci@gmail.com

***Abstract*—We present, to our knowledge, the first systematic evaluation of EmbeddingGemma 2, a 744M-parameter open multimodal embedding model, on African languages. On text we re-run AfriMTEB-Lite for both EmbeddingGemma generations and five baselines under one harness and find that the EmbeddingGemma row in the AfriMTEB paper (29.1) does not reproduce: under the model's intended bidirectional attention the macro score is about 45 for both generations, which share a tokenizer and score within a point of each other, and running the model with a causal instead of a bidirectional attention mask reproduces the published macro (30.5 against 29.1 on its 12 tasks, and 35.8 against 36.0 on the seven tasks whose baselines reproduce), which we believe explains the gap. They trail a size-matched mE5-base by 12 points, more than half of it on cross-lingual bitext mining, while EmbeddingGemma 2 stays within 5.5 points of 560M-parameter models on sentiment, hate-speech and emotion classification. On speech, EmbeddingGemma 2 retrieves transcripts from spoken utterances at 0.934 Recall@1 across 21 African languages zero-shot. A LoRA adapter with 5M parameters, trained for one H100-hour on 9,000 speech–transcript pairs from six languages, raises the 14 languages it was evaluated on from 0.938 to 0.997, transfers to seven unseen languages and to held-out sources including a language in neither FLEURS nor the training set, matches a 1B-parameter ASR cascade in a single pass, and, in the three languages tested at that scale, holds at 0.94–0.99 against a quarter-million candidates where the zero-shot model falls to 0.53–0.70. Cross-lingual alignment is the model's consistent weakness: African speech retrieves English text at 0.43 and images retrieve African-language captions at 0.17 versus 0.79 for English. Stacking a text and an audio adapter recovers nine points of the speech-to-English gap. On WAXAL spontaneous speech in five languages the adapter never saw, it lifts zero-shot retrieval from 0.63 to 0.93. We release the harness, all results, the adapter, and four tools built on it: a transcript validator, voice-note and long-form archive search, and a cross-corpus leakage check, all running on a 4 GB laptop GPU.**

***Index Terms*—embeddings, African languages, speech retrieval, multimodal, LoRA, benchmark reproducibility**

I. Introduction

EmbeddingGemma 2 [1] maps text, images, audio and video into one 768-dimensional space and is small enough to run on a phone: 271M parameters for text, 577M with the audio tower. That combination targets exactly the conditions of African-language deployment, where speech often matters more than text and connectivity is poor. Yet the model card names no African language, and the only prior African-language benchmark result for the family, beyond a few MMTEB tasks in its own report, is a 29.1 macro score for EmbeddingGemma v1 [2] in the AfriMTEB paper [3], with bitext and retrieval scores of 11–18 and clustering of 5 that point to an evaluation-configuration problem rather than the model's ability.

We ask three questions. (1) What does EmbeddingGemma actually score on African text, under the same harness as its competitors? (2) Does the audio tower work for African speech? (3) Where does the shared multimodal space hold, where does it break, and how cheaply can the breaks be repaired?

Our contributions are: a corrected AfriMTEB-Lite leaderboard with a per-task reconciliation against the published one (§III); to our knowledge the first speech-retrieval evaluation of EmbeddingGemma 2 on African languages, with a 5M-parameter adapter that raises it to 0.997 on FLEURS, validated on held-out sources, against an ASR cascade and at scale (§IV); and a characterisation of the cross-lingual failure shared by the text, speech and vision paths (§V).

II. Setup

a) Harness.: `afriembed` is a single command-line interface over sentence-transformers 6.1 [4], transformers 5.19 and mteb 2.24 [5]. Task prompts are taken from each model's own prompt catalog, inference is in bf16, and every result is recorded with its prompts, context cap and library versions. Results quoted without a table are in the released results files. Baselines use mteb's registered model definitions so their published instructions are applied.

b) Text.: AfriMTEB-Lite [3]: 13 tasks over nine languages (Amharic, Oromo, Igbo, Yoruba, Hausa, Swahili, Kinyarwanda, Xhosa, Zulu), comprising Flores and NTREX bitext mining [6], [7], Belebele retrieval [8], AfriXNLI [9], MasakhaNEWS [10], KinNews, SIB-200 classification and clustering [11], Injongo intent [12], AfriSenti [13], AfriHate [14] and EmotionAnalysisPlus. We also run AfriMTEB Full (33 text tasks, 59 languages) and CIRAL [15] (English queries, 50k-passage pools in Hausa, Somali, Swahili, Yoruba). Two upstream task definitions needed workarounds (an empty validation split in EmotionAnalysisPlus; four languages missing from mteb's SIB-200 v2 dataset, the paper's SIB200_14Classes); both are recorded per run and applied identically to all models. Baselines are multilingual-e5 small, base and large-instruct [16], AfriE5 [3] and bge-m3 [17].

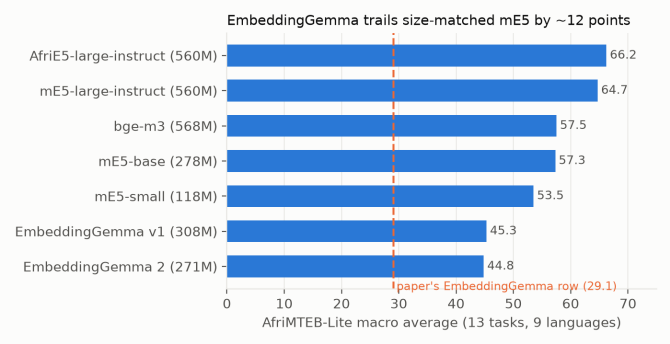


Fig. 1. AfriMTEB-Lite macro score for seven models under one harness. Both EmbeddingGemma generations score about 45 on our 13-task macro (45.0 for v1 on the paper's 12 tasks), not the 29.1 published for v1.

c) Speech.: FLEURS [18] test for 21 African languages (one recording per sentence, pools of about 300); FLEURS n-way parallel ids for cross-lingual speech bitext; AfriSpeech-200 [19] for accented English; FLEURS train for fine-tuning. Held-out sources are BibleTTS [20] and NaijaVoices [21]. Transcripts carry the model's `document` prompt; audio carries no prompt.

d) Vision.: Afri-Aya [22] image–caption pairs with English and local-language captions.

e) Compute.: About 9 H100-hours and 4 A10G-hours on Modal ($42 at list prices), plus a 4 GB laptop GPU for the tools.

III. Text: the corrected benchmark

Table I and Figure 1 give the leaderboard. Three results follow.

a) The published v1 score does not reproduce.: On the paper's own 12 tasks we obtain 45.0 for v1 against the published 29.1, with Belebele 54.9 vs 10.6, Flores 42.1 vs 12.2 and NTREX 49.0 vs 17.9. Table II reconciles every task against the paper. Seven tasks reproduce the paper's baseline numbers for mE5-large-instruct, AfriE5 and bge-m3 within 1.3 points (Belebele, AfriXNLI, Injongo, SIB-200 classification and clustering, AfriHate, and News once MasakhaNEWS and KinNews are merged as the paper does), so for those tasks our harness matches the paper's to within 1.3 points. On those seven tasks v1 differs from the paper by $+3$ to $+44$. Three tasks (Flores, NTREX, AfriSenti) run 2–7 points higher for every baseline, and Emotion 1.4–1.9 lower, consistent with differences between the task definitions the paper used and those in mteb 2.24; the correction claim does not rest on them. Disabling task prompts costs only 1–2 points and running at the model's full 2K context moves the macro by 0.1, so neither explains the published numbers. The paper does not state the framework or library versions, prompts, precision or sequence length used to evaluate EmbeddingGemma. We reproduce the published macro by running the model with a causal (left-to-right) attention mask instead of its bidirectional one: the paper's 12-task macro becomes 30.5 against the published 29.1 (32.0 vs 31.6 without the non-comparable SIB-200 v2), with Flores 14.0 vs 12.2, NTREX 20.1 vs 17.9, SIB-200 classification 29.5 vs 29.6 and AfriHate 46.4 vs 45.4 (Appendix B). On the seven tasks whose baselines reproduce, the causal run gives 35.8 against the published 36.0; the largest per-task gaps are Injongo (41.8 vs 49.4) and Belebele (16.4 vs 10.6). EmbeddingGemma's config enables bidirectional attention through a flag that transformers versions before 4.57 silently ignore, so the encoder runs causally with no warning. This failure is already documented for other benchmarks [23], [24]; what we add is evidence that it is the likely cause of the AfriMTEB row. We have shared these findings with the AfriMTEB authors, whose own limitations section notes that some models may be sensitive to settings they did not exhaustively tune; they have not yet confirmed their original setup. Precision (fp32 45.0; fp16 yields NaN for every input), a 128-token limit (44.1) and dropping the projection layers (under 2 points on two spot-checked tasks) come nowhere near 29.

The paper also does not say which Flores and NTREX language pairs it mines. mteb's Lite definition mines only African–African pairs. Re-running with English pairs included (Appendix A) raises every model by 1–12 points and still does not reproduce the paper's baseline bitext numbers, which sit 2–14 points below ours under either pair set. v1 scores 42–53 on Flores under both protocols against the published 12.2.

b) v2 equals v1 on African text.: 44.8 vs 45.3 on Lite and 52.0 vs 52.6 on the Full suite. Tokenizer fertility is identical to the decimal across 30 languages, and both vocabularies have 262,144 entries, indicating the tokenizer did not change between generations.

c) The gap to mE5 is concentrated.: For EmbeddingGemma 2 against mE5-base, bitext mining is 40–47 vs 89–91 and clustering 14 vs 20 (44 for mE5-large-instruct), while sentiment, hate speech and emotion are within 5.5 points of the 560M-parameter models and news is within 2 of bge-m3.

d) Compression.: Matryoshka truncation [25] retains 0.99 / 0.97 / 0.90 of the macro at 512 / 256 / 128 dimensions. Trimming the vocabulary to the 23k tokens used by the nine languages plus English removes 245 MB in bf16 but costs five macro points.

IV. Speech: zero-shot and adapted

a) Zero-shot.: Speech-to-transcript R@1 averages 0.934 over 21 languages (Figure 2 shows the 14 also evaluated with the adapter). Resource tier barely predicts the outcome (mean 0.94 for the 11 low-resource languages, 0.91 for the 7 mid and 0.97 for the 3 high): Wolof 0.962, Luo 0.974, Fulfulde 0.964 and Kabuverdianu 0.985 are low-resource, while the weakest are Igbo 0.786, Umbundu 0.806, Somali 0.871, Kamba 0.879 and Amharic 0.882. Removing the `document` prompt from transcripts costs up to 23 points, most for Somali (23) and Igbo (13). On African-accented English (AfriSpeech-200, 108 accents, 1,967 utterances, full pool) R@1 is 0.888 with no clinical-versus-general penalty.

TABLE I

AFRIMTEB-LITE (13 TASKS, 9 LANGUAGES). FLORES AND NTREX ARE BITEXT F1; BELEBELE IS NDCG@10; NEWS IS MASAKHANEWS; SIB IS SIB-200 CLASSIFICATION; THE REMAINING COLUMNS ARE CLASSIFICATION ACCURACY. MACRO IS THE MEAN OVER ALL 13 TASKS, INCLUDING KINNEWS, SIB-200 CLUSTERING AND SIB-200 V2, WHICH ARE NOT SHOWN. EMBEDDINGGEMMA ROWS ARE AT EACH MODEL'S FULL CONTEXT (2K FOR V1, 8K FOR V2), OTHER MODELS AT A 1K-TOKEN CAP (512, THEIR MAXIMUM, FOR THE E5 MODELS); AT 1K THE EMBEDDINGGEMMA MACROS ARE 0.1 HIGHER.

Model	Params	Macro	Flores	NTREX	Belebele	AfriXNLI	News	SIB	Intent	Senti	Hate	Emotion
AfriE5-large-instruct	560M	**66.2**	97.9	94.8	76.4	68.8	83.5	71.9	75.0	55.1	50.9	31.3
mE5-large-instruct	560M	64.7	97.8	94.4	76.0	64.2	82.9	71.0	75.5	49.7	51.1	29.6
bge-m3	568M	57.5	85.0	82.8	69.9	68.3	76.8	55.6	74.8	51.2	49.5	28.0
mE5-base	278M	57.3	91.3	88.6	62.9	63.5	74.9	54.9	75.0	51.3	49.0	29.3
mE5-small	118M	53.5	82.8	81.7	53.0	62.1	73.4	52.5	65.0	46.9	47.2	28.0
EmbeddingGemma v1	308M	45.3	42.1	49.0	54.9	59.2	75.0	44.2	59.5	44.2	48.1	26.5
EmbeddingGemma 2	271M	44.8	39.5	47.2	51.8	59.7	75.0	43.6	57.0	49.7	47.9	28.0

TABLE II

OUR SCORE MINUS THE AFRIMTEB PAPER'S, PER TASK (EG V1 IS EMBEDDINGGEMMA V1; ME5-L IS ME5-LARGE-INSTRUCT). NEWS MERGES MASAKHANEWS AND KINNEWS OVER LANGUAGES, AS THE PAPER DOES. THE TOP BLOCK REPRODUCES THE PAPER'S BASELINES WITHIN 1.3; IN THE BOTTOM BLOCK EVERY BASELINE MOVES THE SAME WAY. THE SIB-200 V2 TASK IS OMITTED BECAUSE FOUR OF ITS LANGUAGES ARE MISSING UPSTREAM.

Task	mE5-L	AfriE5	bge-m3	EG v1
AfriHate	−0.4	−0.8	−0.6	+2.7
AfriXNLI	−0.3	−0.2	−0.4	+5.6
Belebele	+0.3	−1.3	−0.0	**+44.3**
Injongo	+0.0	−0.4	−0.6	+10.1
SIB-200 cls.	−0.2	−0.1	+0.0	+14.6
SIB-200 clust.	−0.2	−0.0	+0.1	+12.0
News	−0.4	−0.3	−0.3	+10.8
Emotion	−1.9	−1.5	−1.4	−0.1
AfriSenti	+2.7	+4.4	+3.3	+5.2
NTREX	+2.9	+2.8	+2.4	**+31.1**
Flores	+6.4	+6.7	+6.9	**+29.9**

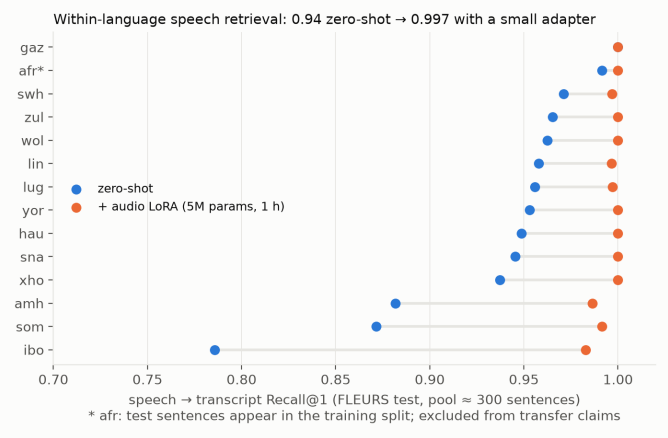


Fig. 2. Speech-to-transcript R@1 per language on FLEURS, zero-shot and with the audio adapter.

b) Adapter.: We train a rank-16 LoRA [26], [27] on the text tower's attention and MLP projections (5.0M parameters, 10 MB on disk) with a cached contrastive loss [28] on 9,000 FLEURS-train speech–transcript pairs from Hausa, Yoruba, Swahili, Amharic, Igbo and Somali for one epoch, about one H100-hour. The mean over the 14 evaluated languages (the six training languages, the seven unseen ones below, and Afrikaans) rises from 0.938 to 0.997 (0.996 without Afrikaans); retraining with two more seeds, which change the LoRA initialisation and data order, gives 0.9964–0.9965, with 128-dimensional retention of 0.988–0.989. The LoRA sits on the text tower, which also processes the audio tower's output, so it is active for both audio and transcripts. Gains concentrate where the base was weakest (Igbo +20, Amharic +10, Somali +12) and transfer to seven languages absent from training (Xhosa, Zulu, Wolof, Luganda, Lingala, Shona and Oromo reach 0.99–1.00; Oromo has only 36 test clips). Training and test sentences are disjoint across languages: FLEURS shares sentence ids between languages, so we checked the test sentences of all 21 languages, the six training languages included, against the 1,509 FLORES sentence ids in the training data, and found no overlap in 20 of them. The exception is Afrikaans, whose 233 test sentences all occur in the other languages' training split, so it is excluded from every transfer claim (script in the released code). Unfreezing the 306M-parameter audio tower adds 0.001. The adapter also repairs low-dimensional embeddings: 128-dimensional retention rises from 0.82 to 0.99 (Figure 3). Evaluating with transcripts shuffled, so that no clip is paired with its own text, gives 0.003 for both models, exactly chance, so the metric does not leak targets.

c) Held-out sources.: To reduce the risk of FLEURS leakage we evaluate on sources outside FLEURS and outside the adapter's training data (Google has not published the base model's training data) (Table III, Figure 4): BibleTTS verses in Hausa and in Ewe, a language in neither FLEURS nor the adapter's training set, and NaijaVoices crowd recordings of about 3 s in Hausa, Igbo and Yoruba, with pools of 409–1,000 against about 300 on FLEURS. The zero-shot model scores 0.60–0.88 and the adapter 0.84–0.995, including 0.995 on Ewe. WAXAL [29], released in February 2026, adds spontaneous, image-prompted speech in five languages the adapter never saw (four of them also in FLEURS): zero-shot R@1 is 0.49–0.76 (mean 0.63) and the adapter raises every language by 24–

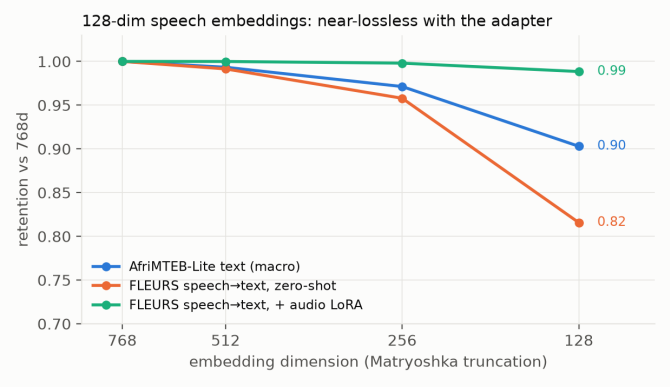


Fig. 3. Retention of full-dimension R@1 under Matryoshka truncation, zero-shot and with the adapter.

TABLE III
SPEECH-TO-TRANSCRIPT R@1 ON SOURCES OUTSIDE FLEURS. EWE IS IN NEITHER FLEURS NOR THE ADAPTER'S TRAINING LANGUAGES; THE FIVE WAXAL LANGUAGES (SPONTANEOUS IMAGE-PROMPTED SPEECH, 17–24 S CLIPS) ARE ALL ABSENT FROM THE ADAPTER'S TRAINING SET.

Source	Lang	Pool	Zero-shot	+LoRA
BibleTTS	hau	998	0.861	0.994
BibleTTS	ewe	409	0.883	0.995
NaijaVoices	hau	1000	0.823	0.959
NaijaVoices	ibo	998	0.599	0.837
NaijaVoices	yor	980	0.765	0.913
WAXAL	lug	600	0.705	0.995
WAXAL	orm	600	0.757	0.993
WAXAL	sna	600	0.645	0.985
WAXAL	lin	600	0.547	0.875
WAXAL	ewe	600	0.488	0.783

34 points (mean 0.93; 0.926–0.928 over three training seeds, Ewe 0.782–0.787), with Luganda, Oromo and Shona above 0.98.

d) Cascade baseline.: The natural alternative to direct speech retrieval is a cascade: transcribe, then embed the transcript. Table IV compares a cascade of MMS-1B-all [30] with greedy decoding and the same text embedder against the direct path over ten languages. Retrieval from noisy transcripts is robust even at 54% word error rate: the cascade reaches 0.990 and beats the zero-shot direct path by 7 points on average and by 20 on Igbo. The adapted direct path, at 0.995, matches or beats the cascade in nine of ten languages (Amharic 0.986 against 1.000) in one pass, with no 1B-parameter recogniser and no transcription step. Its pools are the full test sets (239–350), never smaller than the cascade's, which can only lower its score.

e) Scale.: Pools of 300 flatter any retriever. We grow the candidate set from the queries' own transcripts to all FLEURS sentences, 50k CIRAL news passages, and those passages split into sentences, reaching 235k–273k same-language candidates (Table V, Figure 5). Against the largest pools the adapted model retrieves at 0.967 (Hausa), 0.943 (Yoruba) and 0.987 (Swahili), where the zero-shot model falls to 0.70, 0.53 and 0.70. At 256 dimensions the adapter still scores 0.953 / 0.918 / 0.967. Crowd-recorded NaijaVoices queries at the largest pool move from 0.76 to 0.94 (Hausa) and 0.48 to 0.89 (Yoruba).

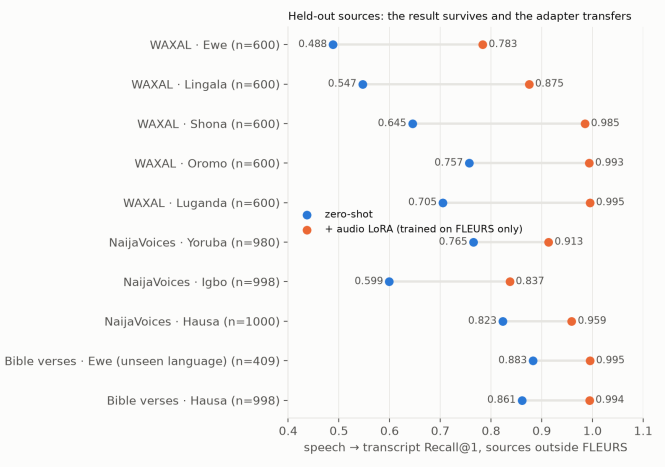


Fig. 4. Held-out sources: zero-shot versus adapted R@1.

TABLE IV
CASCADE (MMS-1B-ALL ASR THEN TEXT EMBEDDING) VERSUS DIRECT SPEECH RETRIEVAL, R@1 ON FLEURS POOLS OF ABOUT 300. DIRECT+LORA USES THE FULL TEST POOL PER LANGUAGE (239–350).

Language	WER	Cascade	Direct	Direct+LoRA
Hausa	0.256	0.993	0.953	1.000
Yoruba	0.535	0.997	0.957	1.000
Swahili	0.147	0.997	0.970	0.997
Amharic	0.273	1.000	0.882	0.986
Igbo	0.442	0.977	0.777	0.983
Somali	0.440	0.980	0.877	0.991
Xhosa	0.370	0.987	0.943	1.000
Zulu	0.337	0.993	0.963	1.000
Wolof	0.388	0.996	0.962	1.000
Luganda	0.439	0.983	0.957	0.997
Mean	0.363	0.990	0.924	**0.995**

V. WHERE THE SHARED SPACE BREAKS: CROSS-LINGUAL ALIGNMENT

Over 13 languages, recordings that retrieve their own transcript at 0.93 retrieve the English translation at only 0.43, and English speech at 0.41 (Figure 6). Only Swahili (0.72) and Amharic (0.63) are well aligned; most languages sit at 0.33–0.45 and African–African pairs average 0.34. The failure appears to lie in the text space: text-to-text bitext is 0.60 on the same pairs, Afri-Aya images retrieve their English caption at 0.79 but their local caption at 0.17, and local and English captions of the same image match at only 0.07–0.19 for nine of the twelve Afri-Aya languages (the others: Igbo 0.38, Egyptian Arabic 0.96, Krio 0.84; pools of about 300). On CIRAL passage retrieval, by contrast, EmbeddingGemma 2 (0.380 nDCG@20) ties mE5-base (0.379), suggesting topical overlap suffices for passages where exact alignment does not for sentences.

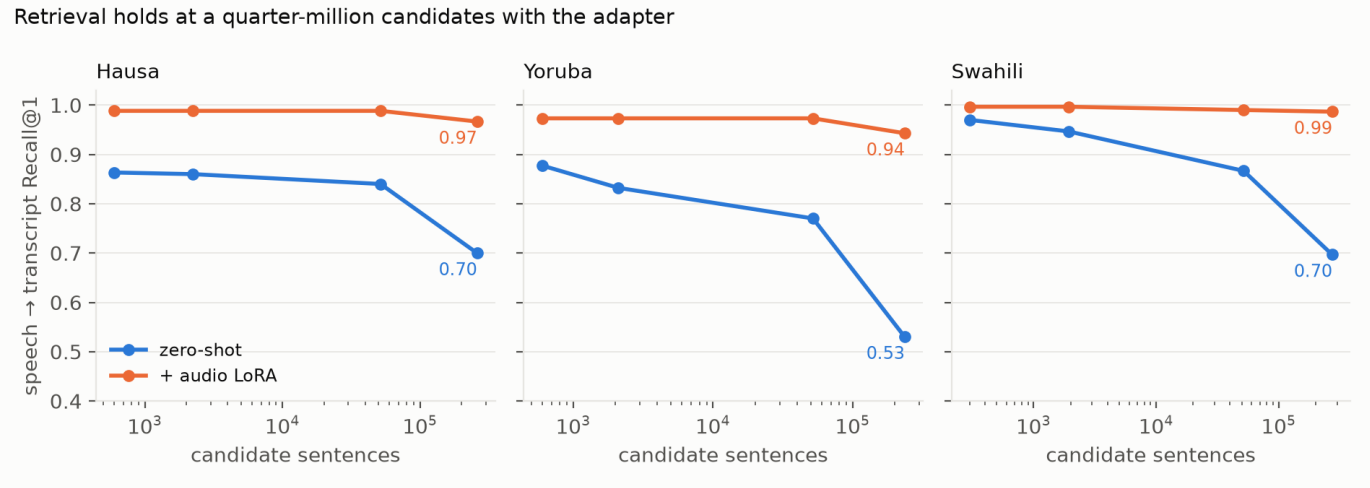


Fig. 5. R@1 against pool size for Hausa, Yoruba and Swahili, zero-shot and with the audio adapter.

TABLE V
R@1 AGAINST NESTED CANDIDATE POOLS (768 DIMENSIONS). THE LARGEST POOLS HOLD 235K–273K SENTENCES. QUERIES ARE 300 FLEURS CLIPS PLUS, FOR HAUSA AND YORUBA, ABOUT 300 NAIJAVOICES CLIPS, SO THE SMALLEST POOL (THE QUERIES' OWN TRANSCRIPTS) HOLDS 300–600 SENTENCES.

Lang	Model	300–600	∼2k	∼52k	∼250k
Hausa	zero-shot	0.863	0.860	0.840	0.700
	+LoRA	0.988	0.988	0.988	**0.967**
Yoruba	zero-shot	0.878	0.832	0.770	0.530
	+LoRA	0.973	0.973	0.973	**0.943**
Swahili	zero-shot	0.970	0.947	0.867	0.697
	+LoRA	0.997	0.997	0.990	**0.987**

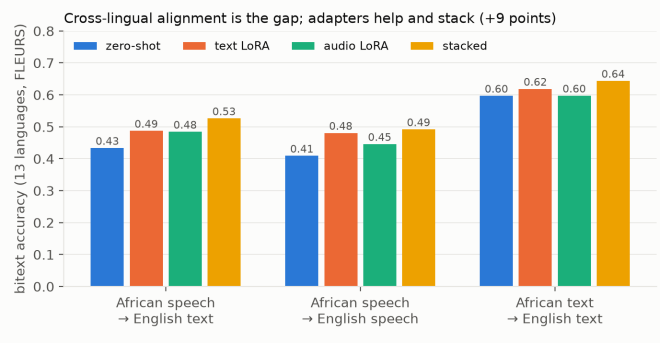


Fig. 6. Cross-lingual bitext R@1, mean over 13 African languages, for the zero-shot model and each adapter configuration.

Two cheap interventions each recover about five points of cross-lingual speech retrieval (Table VI): the audio adapter (0.434 to 0.484) and a text-side LoRA trained on 45k caption translation pairs (0.434 to 0.487). Activating both adapters together is nearly additive (0.526, a 9.2-point gain against 10.3 for the two separately; text-to-text 0.644), with every one of the 13 languages improving. The audio adapter leaves mean text-to-text bitext at 0.597, but per-language scores move

TABLE VI
CROSS-LINGUAL BITEXT R@1 (FLEURS N-WAY PARALLEL IDS, 13 LANGUAGES TO ENGLISH) WITH THE TEXT LORA, THE AUDIO LORA, AND BOTH STACKED.

Direction	Base	+text	+audio	both
audio → English text	0.434	0.487	0.484	**0.526**
audio → English audio	0.409	0.481	0.446	**0.492**
text → English text	0.597	0.619	0.597	**0.644**
text → English audio	0.554	0.573	0.546	**0.602**

both ways (Oromo 0.39 to 0.47, Yoruba 0.55 to 0.52). The text adapter does not improve AfriMTEB-Lite (44.9 to 44.0 at the same 1K-token context) and hurts intent classification, suggesting caption pairs are too small and too narrow a signal. A fine-tune on real parallel corpora is the natural next step, and because audio inherits the text space, it should lift speech cross-lingually as well.

VI. OTHER FINDINGS

Spoken language identification from raw audio embeddings reaches 0.67 5-NN accuracy over 21 languages (chance 0.048), with Zulu/Xhosa the dominant confusion. The Gemma tokenizer spends 1.95× as many tokens per word on African languages as on English, against 1.66× for XLM-R [31] and 1.33× for NLLB-200 [6]; Zulu, Xhosa and Tigrinya cost 2.6–2.7×. Fragmentation does not predict bitext quality (Pearson $r = -0.21$ against text bitext over 13 languages).

VII. TOOLS BUILT ON THE ADAPTER

Four tools follow directly from within-language speech retrieval and run on a 4 GB laptop GPU; each is released with the harness and measured in the results file. (1) A *transcript validator* flags clips whose audio does not match the claimed transcript: with 10% of labels deliberately shuffled, it caught 251 of 251 wrong labels across NaijaVoices Hausa, BibleTTS Hausa and WAXAL Luganda as hard fails at 0.93–1.00 precision (at most 0.9% of clean clips failed; counting borderline

clips sent for review, 1–8% are flagged), and surfaced a genuine label error in NaijaVoices. (2) *Voice-note search*: a recording of a sentence retrieves another reader's recording of the same sentence at 0.996 R@1 over six languages (FLEURS, pools of 175–300). (3) *Long-form archive search*: over 23 minutes of continuous Hausa speech indexed as 10 s windows, a spoken query finds the right window at 0.81 (top-1) and 0.98 (top-3). (4) A *cross-corpus leakage check* finds sentences of one corpus spoken in another from audio alone, with precision 1.00 and recall 0.95 on a positive control and no false positives across three Hausa corpora. A spoken-question benchmark over English passages (Spoken AfriQA) and ASR-free spoken intent routing (Spoken Injongo), both with synthetic MMS-TTS queries, are also released; the former is a negative result in the African-to-English direction (0.19 R@1 spoken vs 0.82 typed in English; the Igbo TTS voice is largely unintelligible: its clips retrieve their own text at only 0.28 and Igbo scores 0.05), the latter reaches 0.80 accuracy over 40 intents for spoken Hausa against 0.92 typed.

VIII. Limitations

FLEURS sentence ids are shared across languages, so we checked every test sentence in all 21 languages against the six training languages' train split: only Afrikaans overlaps (all 233 test sentences), and it is excluded from every transfer claim; the means move by 0.0003 (adapted) and 0.003 (zero-shot) without it. FLEURS is public and may be in the audio encoder's training data; the held-out evaluation on NaijaVoices and BibleTTS reduces this risk, though its pools are still around 1,000. Most retrieval pools are small (about 300); the large-pool test covers three languages up to 273k candidates and the others are untested at that scale. The cascade baseline uses one recogniser with greedy decoding; a stronger recogniser or language-model rescoring would raise it. Baselines re-run through mteb score about 3 points above the paper on its 12-task macro, mostly from SIB-200 v2 (four languages missing upstream) and from Flores, NTREX and AfriSenti; rankings are unaffected, and the correction claim rests only on the seven tasks whose baselines reproduce. The AfriMTEB paper's bitext protocol could not be recovered from its text, so its bitext columns are compared under two pair sets rather than one. Two AfriMTEB-Lite languages (Kinyarwanda, Oromo) have little or no FLEURS audio. The adapter was trained with three seeds; every other result comes from a single run.

IX. Conclusion

EmbeddingGemma 2 is a weak cross-lingual aligner for African languages and a middling within-language text embedder, but its audio tower works for African speech out of the box. For on-device voice search in a single language the zero-shot model is usable but is beaten by a conventional recogniser-plus-embedder cascade; a 10 MB adapter closes that gap in a single pass, is near-perfect on FLEURS, including seven languages it never saw, reaches 0.78–0.995 on held-out recordings, and, in the three languages tested, holds up at indexes of a quarter-million sentences. Cross-lingual use needs a real parallel-data fine-tune, which the stacking result suggests will carry over to speech.

TABLE VII
Bitext F1 with English–African pairs (eng) and African–African pairs (Afr) against the AfriMTEB paper's figures. The Afr columns come from this rerun and differ from Table I by at most 0.1.

	Flores		NTREX		Paper	
Model	eng	Afr	eng	Afr	Flores	NTREX
EmbeddingGemma v1	53.2	42.2	52.5	49.0	12.2	17.9
EmbeddingGemma 2	51.4	39.5	49.6	47.2	–	–
mE5-large-instruct	99.2	97.9	98.1	94.3	91.4	91.5
AfriE5	99.3	97.9	98.1	94.8	91.2	92.0
bge-m3	91.6	85.0	88.9	82.8	78.1	80.4

Artifacts

The harness, raw per-run results with prompts, versions and caveats, and all figures are released with the paper. The audio adapter is available at https://huggingface.co/naijaml/embeddinggemma-2-african-speech-lora.

Appendix A
Bitext mining with English pairs

mteb's AfriMTEB-Lite definition mines the 72 African–African Flores pairs among the nine Lite languages and 30 NTREX pairs among the six it covers, with no English (Kinyarwanda appears in NTREX only paired with English). The AfriMTEB paper does not state which pairs it mines. Table VII reruns both tasks with English pairs (18 for Flores, 14 for NTREX).

English pairs score higher than African–African pairs for every model, and neither pair set reproduces the paper's baseline bitext numbers.

Appendix B
Reproducing the published score

EmbeddingGemma v1 on AfriMTEB-Lite under the same harness, with its bidirectional attention (correct) and with a causal mask, against the AfriMTEB paper's published row.

Under a causal mask, sentiment, hate-speech, emotion and NLI lose only 1–4 points, topic and intent classification lose 13–18, and bitext, retrieval and clustering collapse by 13–39, which is the shape of the published row. The Correct column is the full-context run used throughout; the causal run used a 1K-token cap, at which the correct model gives News 70.0 and a 12-task macro of 45.1, so the cap does not affect the comparison. ‡MasakhaNEWS and KinNews merged and averaged over languages, as the paper does. †Not comparable: mteb's current SIB-200 v2 (the paper's SIB200_14Classes) lacks four Lite languages and every baseline runs about 25 points above the paper on it. AfriSenti runs 4 points above the paper here, as it does (3–4 points) for every baseline (Table II).

TABLE VIII
EMBEDDINGGEMMA V1 WITH CORRECT (BIDIRECTIONAL) AND CAUSAL ATTENTION AGAINST THE AFRIMTEB PAPER'S ROW.

Task	Correct	Causal	Paper
SIB-200 cls.	44.2	29.5	29.6
AfriHate	48.1	46.4	45.4
Emotion	26.5	24.8	26.6
AfriXNLI	59.2	55.6	53.6
SIB-200 clust.	16.6	3.4	4.6
Flores	42.1	14.0	12.2
NTREX	49.0	20.1	17.9
Belebele	54.9	16.4	10.6
Injongo	59.5	41.8	49.4
News[‡]	69.8	57.2	59.0
SIB-200 v2[†]	26.2	14.2	1.4
AfriSenti	44.2	43.0	39.0
Macro (12 tasks)	**45.0**	**30.5**	**29.1**
Macro (11, no v2)	46.7	32.0	31.6

TABLE IX
MACRO SCORES UNDER THREE AGGREGATIONS (OUR 13 TASKS; THE PAPER'S 12 WITH NEWS MERGED; 11 WITH SIB-200 V2 DROPPED) AND THE PAPER'S PUBLISHED FIGURES.

Model	13	12 (paper)	11	Paper
EmbeddingGemma v1	45.3	45.0	46.7	29.1
EmbeddingGemma 2	44.8	44.4	46.2	–
mE5-large-instruct	64.7	65.0	66.5	62.0
AfriE5	66.2	66.6	67.9	63.7
bge-m3	57.5	57.8	59.9	55.0

APPENDIX C
AGGREGATING THE PAPER'S WAY

The paper's AfriMTEB-Lite is 12 tasks: MasakhaNEWS and KinNews are one News task, and its SIB-200 task covers all nine Lite languages. Table IX re-aggregates our runs under that scheme and with the incomparable SIB-200 v2 task dropped. Merging the news tasks moves every macro by under 0.5; dropping SIB-200 v2 raises each by a further 1.3–2.1.

APPENDIX D
ADAPTER TRAINING DETAILS

LoRA rank 16, alpha 32, no dropout, applied to the query, key, value, output, gate, up and down projections of the text tower only (5.0M trainable parameters, PEFT 0.21). Cached multiple-negatives ranking loss on the 768-dimensional embedding, batch size 32, learning rate 2×10^{-5} with 10% warmup, one epoch over 9,000 FLEURS-train pairs from Hausa, Yoruba, Swahili, Amharic, Igbo and Somali, bf16, one H100 (sentence-transformers 6.1, transformers 5.19). The adapter is loaded with sentence-transformers' `load_adapter` and is 10 MB on disk. Evaluating with shuffled transcripts gives 0.003 R@1, exactly chance.

REFERENCES

[1] Google DeepMind, "EmbeddingGemma 2 model card," https://huggingface.co/google/embeddinggemma-2, 2026, released 2026-10-06, Apache 2.0.

[2] H. Schechter Vera, S. Dua, B. Zhang, D. Salz *et al.*, "EmbeddingGemma: Powerful and lightweight text representations," 2025.

[3] K. Uemura, M. Zhang, and D. I. Adelani, "AfriMTEB and AfriE5: Benchmarking and adapting text embedding models for African languages," in *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2026, pp. 3697–3717.

[4] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, 2019, pp. 3982–3992.

[5] N. Muennighoff, N. Tazi, L. Magne, and N. Reimers, "MTEB: Massive text embedding benchmark," in *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, 2023, pp. 2014–2037.

[6] NLLB Team, M. R. Costa-jussà, J. Cross, O. Çelebi, M. Elbayad, K. Heafield *et al.*, "No language left behind: Scaling human-centered machine translation," 2022.

[7] C. Federmann, T. Kocmi, and Y. Xin, "NTREX-128 – news test references for MT evaluation of 128 languages," in *Proceedings of the First Workshop on Scaling Up Multilingual Evaluation*, 2022, pp. 21–24.

[8] L. Bandarkar, D. Liang, B. Muller, M. Artetxe, S. N. Shukla, D. Husa, N. Goyal, A. Krishnan, L. Zettlemoyer, and M. Khabsa, "The Belebele benchmark: a parallel reading comprehension dataset in 122 language variants," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024, pp. 749–775.

[9] D. I. Adelani, J. Ojo, I. A. Azime, J. Y. Zhuang, J. O. Alabi, X. He, M. Ochieng, S. Hooker, A. Bukula, E.-S. A. Lee *et al.*, "IrokoBench: A new benchmark for African languages in the age of large language models," in *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics*, 2025, pp. 2732–2757, introduces AfriXNLI.

[10] D. I. Adelani, M. Masiak, I. A. Azime, J. Alabi, A. L. Tonja, C. Mwase, O. Ogundepo, B. F. P. Dossou *et al.*, "MasakhaNEWS: News topic classification for African languages," in *Proceedings of the 13th International Joint Conference on Natural Language Processing and the 3rd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*, 2023, pp. 144–159.

[11] D. I. Adelani, H. Liu, X. Shen, N. Vassilyev, J. O. Alabi, Y. Mao, H. Gao, and E.-S. A. Lee, "SIB-200: A simple, inclusive, and big evaluation dataset for topic classification in 200+ languages and dialects," in *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics*, 2024, pp. 226–245.

[12] H. Yu, J. O. Alabi, A. Bukula, J. Y. Zhuang, E.-S. A. Lee *et al.*, "INJONGO: A multicultural intent detection and slot-filling dataset for 16 African languages," in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2025.

[13] S. H. Muhammad, I. Abdulmumin, A. A. Ayele, N. Ousidhoum, D. I. Adelani, S. M. Yimam *et al.*, "AfriSenti: A Twitter sentiment analysis benchmark for African languages," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023, pp. 13 968–13 981.

[14] S. H. Muhammad, I. Abdulmumin, A. A. Ayele, D. I. Adelani *et al.*, "AfriHate: A multilingual collection of hate speech and abusive language datasets for African languages," in *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics*, 2025, pp. 1854–1871.

[15] M. Adeyemi, A. Oladipo, X. Zhang, D. Alfonso-Hermelo, M. Rezagholizadeh, B. Chen, A.-H. Omotayo, I. Abdulmumin, N. A. Etori *et al.*, "CIRAL: A test collection for CLIR evaluation in African languages," in *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2024.

[16] L. Wang, N. Yang, X. Huang, L. Yang, R. Majumder, and F. Wei, "Multilingual E5 text embeddings: A technical report," 2024.

[17] J. Chen, S. Xiao, P. Zhang, K. Luo, D. Lian, and Z. Liu, "M3-embedding: Multi-linguality, multi-functionality, multi-granularity text embeddings through self-knowledge distillation," in *Findings of the Association for Computational Linguistics: ACL 2024*, 2024, pp. 2318–2335.

[18] A. Conneau, M. Ma, S. Khanuja, Y. Zhang, V. Axelrod, S. Dalmia, J. Riesa, C. Rivera, and A. Bapna, "FLEURS: Few-shot learning evaluation of universal representations of speech," in *IEEE Spoken Language Technology Workshop (SLT)*, 2023.

[19] T. Olatunji, T. Afonja, A. Yadavalli, C. C. Emezue, S. Singh, B. F. P. Dossou, J. Osuchukwu, S. Osei, A. L. Tonja, N. Etori, and C. Mbataku, "AfriSpeech-200: Pan-African accented speech dataset for clinical and general domain ASR," *Transactions of the Association for Computational Linguistics*, vol. 11, pp. 1669–1685, 2023.

[20] J. Meyer, D. I. Adelani, E. Casanova, A. Öktem, D. Whitenack, J. Weber, S. Kabongo, E. Salesky, I. Orife, C. Leong *et al.*, "BibleTTS: a large, high-fidelity, multilingual, and uniquely African speech corpus," in *Proceedings of Interspeech*, 2022, pp. 2383–2387.

[21] C. Emezue, NaijaVoices Community, B. Awobade, A. T. Owodunni *et al.*, "The NaijaVoices dataset: Cultivating large-scale, high-quality, culturally-rich speech data for African languages," in *Proceedings of Interspeech*, 2025.

[22] Cohere Labs Community, "Afri-Aya," https://huggingface.co/datasets/CohereLabsCommunity/afri-aya, 2025, released December 2025.

[23] MTEB contributors, "Results for google/embeddinggemma-300m on wikipediarerankingmultilingual differ from released numbers," https://github.com/embeddings-benchmark/mteb/issues/3144, 2025, gitHub issue #3144, September 2025.

[24] sentence-transformers contributors, "embeddinggemma-300m: silently uses causal attention on transformers $<$ 4.57, producing wrong embeddings," https://github.com/huggingface/sentence-transformers/issues/3725, 2026, gitHub issue #3725, April 2026.

[25] A. Kusupati, G. Bhatt, A. Rege, M. Wallingford, A. Sinha, V. Ramanujan, W. Howard-Snyder, K. Chen, S. Kakade, P. Jain, and A. Farhadi, "Matryoshka representation learning," in *Advances in Neural Information Processing Systems*, 2022.

[26] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-rank adaptation of large language models," in *International Conference on Learning Representations*, 2022.

[27] S. Mangrulkar, S. Gugger, L. Debut, Y. Belkada, S. Paul, B. Bossan, and M. Tietz, "PEFT: State-of-the-art parameter-efficient fine-tuning methods," https://github.com/huggingface/peft, 2022.

[28] L. Gao, Y. Zhang, J. Han, and J. Callan, "Scaling deep contrastive learning batch size under memory limited setup," in *Proceedings of the 6th Workshop on Representation Learning for NLP*, 2021, pp. 316–321.

[29] A. Diack, P. Nelson, K. Agbesi, A. Nakalembe *et al.*, "WAXAL: A large-scale multilingual African language speech corpus," 2026, data: https://huggingface.co/datasets/google/WaxalNLP.

[30] V. Pratap, A. Tjandra, B. Shi, P. Tomasello, A. Babu, S. Kundu, A. Elkahky, Z. Ni, A. Vyas, M. Fazel-Zarandi, A. Baevski, Y. Adi, X. Zhang, W.-N. Hsu, A. Conneau, and M. Auli, "Scaling speech technology to 1,000+ languages," *Journal of Machine Learning Research*, vol. 25, no. 97, pp. 1–52, 2024.

[31] A. Conneau, K. Khandelwal, N. Goyal, V. Chaudhary, G. Wenzek, F. Guzmán, E. Grave, M. Ott, L. Zettlemoyer, and V. Stoyanov, "Unsupervised cross-lingual representation learning at scale," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 2020, pp. 8440–8451.